SADAT JUNAID AKUNJEE

Melbourne, VIC | +61 8197810946 | sadatjunaid786@gmail.com | linkedin.com/in/sadatjunaid | github.com/Invocate-coder17
Visa: Student Visa (Subclass 500) | Available: Part-time now, Full-time from Dec 2026

PROFESSIONAL SUMMARY

100% win rate in competitive AI benchmarking. 98.9% path completion across 2,832 agents. Live AWS-deployed platform serving real users across three languages, built to close a documented 20-point health screening participation gap affecting CALD Australians. Graduate AI and software engineer who ships end-to-end production systems, from containerised cloud backends and multilingual text classifiers to intelligent planning agents, grounded in peer-reviewed evidence and real-world constraints. Published researcher (IJARSET, 2023). Strong communicator who translates complex technical systems into clear outcomes for cross-functional teams. Completes a Master of Artificial Intelligence at Monash University in Dec 2026; actively seeking graduate and intern roles in AI engineering, software development, and applied machine learning.

CORE COMPETENCIES

Technical Skills: Predictive Modelling | Image Classification | Computer Vision | Agent-based Planning | Data Modelling | Feature Extraction | Unit Testing | RESTful API Design | Cloud Deployment | Containerisation | Machine Learning | Symbolic Planning (PDDL)
AI / ML Frameworks & Libraries: PyTorch | TensorFlow | scikit-learn | OpenCV | NumPy | pandas | scikit-image | SciPy | R
Other Frameworks & Tools: FastAPI | Node.js | Express.js | Vue.js | React | Bootstrap | PostgreSQL | MongoDB | Oracle SQL | PL/SQL | Docker | Git | AWS | Azure (AZ-900 Certified)
Programming Languages: Python | Java | JavaScript | SQL | R

KEY PROJECTS

Screening Pathway Companion | *Live, AWS-Deployed Platform*

- 20-point participation gap closed: shipped a production-grade multilingual health platform live across English, Punjabi, and Chinese targeting CALD Australians aged 45 to 74 (AIHW 2026: 24 to 31% vs 44 to 47% English-speaker rate), accessible without a GP referral or account, removing two of the most cited access barriers documented in the evidence base.
- Architected the complete cloud backend: FastAPI containerised on AWS ECS Fargate, private PostgreSQL on Amazon RDS, ECR image registry, Secrets Manager for credentials, and government open data ingestion pipelines from City of Melbourne and PTV GTFS datasets.
- 16 intent categories classified in real time: built a multilingual text classifier using Python and HuggingFace Transformers with a sentence-embedding quality gate, negation checks, and mandatory bilingual human review before any health content is served, a zero hallucination architecture grounded in 34 verified sources, preventing unsafe health guidance from reaching vulnerable users.
- 68% FIT completion rate (Griffith et al. 2019) informed the usability design: applied teach-back methodology through persona-driven empathy mapping, comprehension checks per kit step, and culturally grounded content review, ensuring the platform works for users with low health literacy and limited English proficiency.
- Technology: Python | FastAPI | HuggingFace Transformers | PostgreSQL | AWS ECS | Docker | Vue.js | Git
  Monash FIT5120 | https://github.com/Invocate-coder17

Hybrid AI Planning and Reasoning System | *Competitive AI Benchmarking Project*

- 100% win rate (10 for 10), average score lifted from 4.5 to 11.4, agent losses reduced by 77% (5.2 to 1.2 per game): outperformed a staff-built AI baseline as sole implementer by engineering a four-stage hybrid AI agent that combined symbolic planning, heuristic pathfinding, and adaptive decision-making, each stage measurably improving on the last.
- Delivered a complete hybrid architecture progressing from a distance heuristic baseline to a PDDL symbolic planner, danger-aware A* pathfinding, and a full agent decision-making layer with shared inter-agent coordination memory across all four stages.
- Technology: Python | PDDL | A* search | Multi-agent planning
  Monash FIT5222 | https://github.com/Invocate-coder17

Computer Vision Pipeline | *Image Analysis and Stitching Project*

- 82 to 99% benchmark overlap with production skimage library: implemented a full image stitching pipeline from scratch, Harris corner detection, homography estimation via SVD, RANSAC (2,000 iterations, 5px inlier threshold), and SIFT-based feature matching, with zero use of black-box library functions for any core component, validating mathematical correctness at every stage.
- Technology: Python | NumPy | OpenCV | scikit-image
  Monash FIT5221 | https://github.com/Invocate-coder17

Relational Database and Document Store Design | *Transport Management System Project*

- Cross-paradigm data modelling across two database systems: designed a full Oracle SQL schema with FK constraints, check constraints, composite unique keys, window function queries, and LISTAGG-based JSON document generation,

then mirrored the same architecture in MongoDB using insertMany with nested embedded subdocuments, demonstrating ability to work across both relational and NoSQL data layers in a single system.

- Technology: Oracle SQL | PL/SQL | MongoDB | JSON

Monash FIT9132 | *https://github.com/Invocate-coder17*

FourShare | *Hackathon Project*

- Judge-recognised at university level: led a team from zero to live pitch within a hackathon window, scoped the problem, divided work, and delivered a full-stack React and MongoDB traveller companion app helping people find essential services in a new city, demonstrating ownership, fast execution, and cross-functional team leadership under time pressure.
- Technology: React | MongoDB | JavaScript | Bootstrap | HTML5 | CSS3

Hackathon Project, Reva University | *https://github.com/Invocate-coder17*

WORK EXPERIENCE

Marketing Manager | *MOSAIC: Monash Students for AI with Communities* | Jul 2025 – Present

- 140+ students and stakeholders reached: led all marketing and communications for a Monash AI-for-social-impact organisation, driving visibility for real-world AI projects built for partners including the International Federation of Red Cross (IFRC), a secondary homelessness support application (Nestle-In), and an aged care co-design tool, contributing to packaging these projects as scalable AI toolkits for global distribution through Oxfam and IFRC.
- Built brand presence and audience across social platforms by designing and executing multi-channel campaigns; contributed to packaging AI toolkits for global distribution through Oxfam and IFRC.

College Advisor | *Monash Student Communities* | Jan 2025 – Feb 2026

- 70+ postgraduate and international students supported: guided students through academic pathways, unit selection, and university navigation at Monash, drawing on direct experience as an international student to provide culturally grounded, empathetic one-on-one support across a diverse cohort, demonstrating stakeholder engagement, interpersonal communication, and initiative beyond academic requirements.

Web Developer | *GMAT Verzeo* | Aug 2020 – Sep 2020

- Delivered a client-facing responsive front-end web project within a fixed deadline as part of a remote team; applied HTML, CSS, and JavaScript to build a functional interface and presented completed work to stakeholders.

EDUCATION

Master of Artificial Intelligence | *Monash University, Melbourne* | Dec 2026

B.Tech Computer Science Engineering | *Reva University, Bangalore* | 2019 – 2023

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Azure Fundamentals AZ-900
- Google Analytics for Beginners (Google Certified)
- Web Development Certificate
- Computer Network Fundamentals Certificate
- Published Research Paper: "Product Classification using Deep Learning", IJARSET, Vol. 10, Issue 4, April 2023 (ISSN: 2350-0328, pp. 20602 to 20610); CNN/ResNet-18 pipeline achieving 0.9 training accuracy across 7 product classes, 5,025 images
- Marketing Manager, MOSAIC: Monash Students for AI with Communities, 2025 to Present
- College Advisor, Monash Student Communities, 2025 to 2026

REFERENCES

Vikrant Parmar | Staff, Monash University | Rana.Parmar@monash.edu | Reference for FIT5120 Industry Experience Project and technical mentorship

Adam Fernandes | Staff, Monash University | adam.fernandes@monash.edu | +61 404710902 | Reference for Monash Student Communities

Victoria Ignateva | Staff, Monash University | Victoria.Ignateva@monash.edu | +61 491666007 | Reference for MOSAIC and academic projects

Madhumitha Challa | Casual Staff, Student, IT Postgraduate Peer Mentor, Monash University | rchmadhumitha@gmail.com | +61 468934292